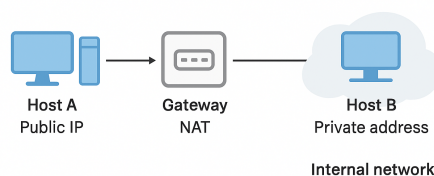




Departamento de Engenharia Informática
Segurança em Tecnologias da Informação, MEI 2024/2025
Exam – Regular Period (1h45)
9 Jun 2025

1. Consider that Ana wants to send a secure message to Luís using the RSA algorithm. The message should be encrypted in such a way that only Luís can decrypt it. Luís chooses the prime numbers $p=5$ and $q=7$.
 - a) Demonstrate how the RSA algorithm can be used to send a message M represented by the number “2” to Luís.
 - b) Demonstrate how Luís can recover the original message. Note: If you are unable to perform all the calculations, just indicate them.
2. The company Acme SA uses the DNS domain “acme.pt” and has been receiving complaints about email spoofing from customers who are receiving fraudulent messages that appear to come from the company’s domain. Assume that the company has not yet implemented DKIM and DMARC. Describe how these mechanisms could be used to protect the domain against this type of attack. Justify your answer.
3. For the scenario illustrated in the following figure, consider that the goal is to ensure secure communications between host A and host B, using IPSec for this purpose. Note that host A uses a public IP, while host B is in an internal network with private addressing. The gateway implements NAT (Network Address Translation), allowing host B to communicate with the outside.



Explain how the IPSec protocol can be used to ensure the confidentiality, integrity, and authentication of communications. Indicate which mode(s) (transport and/or tunnel) and which security header(s) (AH and/or ESP) are most appropriate for the context. Justify your choice, considering the operation of NAT.

4. Consider the principles of “confusion” and “diffusion” defined by Claude Shannon in the context of symmetric cryptography. Explain how these principles are implemented in modern algorithms (such as AES). What is the role of the “avalanche effect”? Justify your answer.
5. An administrator configured a web server to support secure connections (HTTPS), using a digital certificate issued by an official certification authority (“Very safe CA”). However, when accessing the site via HTTPS, the browser displays a security warning stating that the certificate is not trusted (it does not recognize the issuing authority). The “Very safe CA” authority is an intermediate authority and was signed by a root authority (“Root Trusted CA”) that is already included in the browser. Why does the browser display the security warning? Justify your answer, indicating how the issue can be addressed and whether the solution should be applied on the client or server side.
6. Consider the EAP (Extensible Authentication Protocol), used by IEEE 802.11i in the user authentication process for secure Wi-Fi networks. Explain the role of EAP and its characteristics regarding the flexibility and extensibility of authentication mechanisms.
7. Consider the Kerberos protocol. Suppose an attacker manages to capture messages exchanged during the authentication phase, with the intention of reusing them later to impersonate another user and gain unauthorized access to services. How does this protocol implement security against this type of attack (replay)? Justify your answer.
8. Identify and describe two practical applications, situations, or mechanisms in which it is necessary to combine the use of hash functions with asymmetric encryption. Justify your answer.